

Research Report: Accessing Open WebUI from Outside the Local Network Using Ngrok

Introduction

Open WebUI is a web interface for interacting with AI models running locally through Ollama. By default it is available only on the local machine using localhost:8080. This report explains how to make it accessible from outside the local network using Ngrok.

What is Open WebUI?

Open WebUI provides a browser-based interface with login, chat, model selection, knowledge management and user roles. It allows an administrator to manage users while normal users can interact with AI models.

What is localhost and port 8080?

localhost refers to the current computer. The address <http://localhost:8080> points to a service running on your own machine. Port 8080 is simply a communication endpoint used by the application.

Starting Open WebUI

To start the server, open Command Prompt and run:

```
open-webui serve
```

Then open <http://localhost:8080> in a browser. If the server is not running, the page will not load.

Ollama and Models

Ollama runs local large language models. Example commands:

```
ollama pull llama3.2
ollama pull mistral
ollama pull deepseek-r1:8b
```

Use 'ollama list' to view installed models.

Admin and User Roles

In this project Lakshman acts as Admin while Nandita and Vamshith are Users. The admin can manage accounts and configuration. Users can log in and chat with AI according to their permissions.

Problem with Localhost

A remote user cannot access another person's localhost. If Vamshith is in another city and types <http://localhost:8080>, he reaches his own computer rather than Lakshman's.

Ngrok

Ngrok creates a secure tunnel from the internet to a local application. After installation and authentication, it provides a public HTTPS URL that forwards traffic to localhost:8080.

Installing Ngrok

1. Create a free Ngrok account.
2. Install Ngrok.
3. Verify installation with:

```
ngrok version
```

4. Add the authentication token:

```
ngrok config add-authtoken YOUR_AUTHTOKEN
```

Creating a Public URL

Start Open WebUI:

```
open-webui serve
```

In another Command Prompt run:

```
ngrok http 8080
```

Ngrok generates a URL such as <https://example.ngrok-free.app> which forwards to <http://localhost:8080>.

How Remote Access Works

Remote User -> Internet -> Ngrok Public URL -> Ngrok Tunnel -> localhost:8080 -> Open WebUI on the host computer. The host computer must remain powered on with both Open WebUI and Ngrok running.

Advantages and Limitations

Advantages: Easy remote access, HTTPS security, no router configuration required, useful for demonstrations.

Limitations: Tunnel works only while Ngrok is running; the host computer must stay online; access should be protected with user accounts.

Conclusion

Open WebUI is locally hosted software that normally works only on localhost. Ngrok securely exposes it through a temporary public URL, enabling users on different networks to access the application while it continues to run on the host computer.